

APPENDIX
OMITTED PROOFS

Proof of Theorem 3.2. Fix a constant $\psi \geq 1$. We give a gap-preserving reduction from INDEPENDENT SET on 3-regular graphs. Let $(H = (U, F), \ell)$ be an instance, where the question is whether H contains an independent set of size at least ℓ . Write $n_H := |U|$.

Choice of the gap parameter. Choose an even integer Q such that

$$Q > \psi\ell \quad \text{and} \quad Q \geq 8.$$

Since ψ is a fixed constant, Q is polynomial in the input size.

Katz scores. We use the Katz centrality vector

$$s_G = (I - \alpha A(G)^\top)^{-1} \mathbf{1},$$

with $\alpha = \frac{1}{2}$. Since the graph constructed below is a DAG of depth at most 3, all Katz scores can be computed by summing walks of length at most 3.

Construction. We construct a directed, unweighted graph G with two groups *red* and *blue*.

For each vertex $u_i \in U$, create:

- a selector tail t_i , colored *blue*;
- a candidate node r_i , colored *red*.

We make each selector tail have score exactly $Q + 1$ by adding $2Q$ private leaf nodes

$$y_{i,1}, \dots, y_{i,2Q},$$

colored *blue*, with fixed edges

$$y_{i,a} \rightarrow t_i \quad \text{for all } a \in [2Q].$$

Thus

$$s(t_i) = 1 + \alpha(2Q) = Q + 1.$$

For each edge $e = \{u_i, u_j\} \in F$, create ℓ conflict clones

$$q_{e,1}, \dots, q_{e,\ell},$$

colored *blue*, and add fixed edges

$$r_i \rightarrow q_{e,h}, \quad r_j \rightarrow q_{e,h}, \quad \text{for every } h \in [\ell].$$

Next create ℓ blue buffer nodes

$$w_1, \dots, w_\ell.$$

Each buffer node w_a is constructed to have Katz score

$$s(w_a) = \frac{Q}{2} + \frac{3}{4}.$$

To achieve this, add $Q - 2$ private leaves

$$d_{a,1}, \dots, d_{a,Q-2}$$

with fixed edges $d_{a,b} \rightarrow w_a$, and add one intermediate node p_a with one private leaf $z_a \rightarrow p_a$ and one fixed edge $p_a \rightarrow w_a$. Then $s(p_a) = 1 + \alpha = \frac{3}{2}$, and therefore

$$s(w_a) = 1 + \alpha(Q - 2) + \alpha \cdot \frac{3}{2} = 1 + \frac{Q - 2}{2} + \frac{3}{4} = \frac{Q}{2} + \frac{3}{4}.$$

Finally, create ℓ fallback red nodes

$$f_1, \dots, f_\ell,$$

colored *red*. For each fallback node f_a , create Q private blue leaves

$$g_{a,1}, \dots, g_{a,Q}.$$

The edges $g_{a,b} \rightarrow f_a$ will be admissible edits rather than fixed edges.

Admissible edits. The admissible action set $\mathcal{A}$ consists of two types of edges:

- core selector edges

$$t_i \rightarrow r_i \quad \text{for every } u_i \in U;$$

- fallback edges

$$g_{a,b} \rightarrow f_a \quad \text{for every } a \in [\ell], b \in [Q].$$

There are no other admissible edits.

Top- k parameters. Set

$$k := n_H + \ell, \quad \pi := \frac{\ell}{k}, \quad \varepsilon := \frac{1}{4k^2}.$$

Because the number of red nodes in the top- k is an integer, any nonzero deviation from the target red count ℓ is at least $1/k$ in proportion. Hence

$$\text{Unfair}(T_k(G \oplus \Delta)) < \varepsilon$$

implies

$$|T_k(G \oplus \Delta) \cap V_{\text{red}}| = \ell.$$

Thus feasibility for the dual instance is equivalent to achieving exactly ℓ red nodes in the top- k set.

Score levels. Let G' be any graph obtained by adding admissible edits. For a core candidate r_i , there is at most one admissible incoming selector edge $t_i \rightarrow r_i$. Thus:

$$s(r_i) = \begin{cases} 1, & \text{if } t_i \rightarrow r_i \text{ is not added,} \\ 1 + \alpha s(t_i) = 1 + \frac{Q+1}{2} = \frac{Q}{2} + \frac{3}{2}, & \text{if } t_i \rightarrow r_i \text{ is added.} \end{cases}$$

We say that u_i is selected if the edge $t_i \rightarrow r_i$ is added.

Consider a conflict clone $q_{e,h}$ for $e = \{u_i, u_j\}$. If neither endpoint is selected, then

$$s(q_{e,h}) = 1 + \alpha(1 + 1) = 2.$$

If exactly one endpoint is selected, then

$$s(q_{e,h}) = 1 + \alpha\left(\frac{Q}{2} + \frac{3}{2} + 1\right) = \frac{Q}{4} + \frac{9}{4}.$$

If both endpoints are selected, then

$$s(q_{e,h}) = 1 + \alpha\left(\frac{Q}{2} + \frac{3}{2} + \frac{Q}{2} + \frac{3}{2}\right) = \frac{Q}{2} + \frac{5}{2}.$$

For a fallback node f_a , if exactly r of its Q private fallback edges have been added, then

$$s(f_a) = 1 + \frac{r}{2}.$$

In particular, if $r < Q$, then

$$s(f_a) \leq 1 + \frac{Q-1}{2} = \frac{Q}{2} + \frac{1}{2},$$

whereas if all Q private fallback edges are added, then

$$s(f_a) = 1 + \frac{Q}{2} = \frac{Q}{2} + 1.$$

For $Q \geq 8$, the relevant score levels satisfy the strict ordering

$$Q + 1 > \frac{Q}{2} + \frac{5}{2} > \frac{Q}{2} + \frac{3}{2} > \frac{Q}{2} + 1 > \frac{Q}{2} + \frac{3}{4} > \frac{Q}{2} + \frac{1}{2} > \frac{Q}{4} + \frac{9}{4} > 2 > 1.$$

Therefore the n_H selector tails t_i are always the highest-scoring nodes in the graph and hence always belong to the top- k set. The remaining ℓ positions of the top- k set are determined by the nodes below the selector tails.

YES case. Suppose H contains an independent set $S \subseteq U$ with $|S| \geq \ell$. Choose any subset $S' \subseteq S$ of size exactly ℓ . Add the ℓ selector edges

$$\{t_i \rightarrow r_i : u_i \in S'\}.$$

Every selected red candidate r_i has score $\frac{Q}{2} + \frac{3}{2}$. Since S' is independent, every conflict clone has at most one selected endpoint and therefore has score at most

$$\frac{Q}{4} + \frac{9}{4} < \frac{Q}{2} + \frac{3}{4} < \frac{Q}{2} + \frac{3}{2}.$$

The buffer nodes have score $\frac{Q}{2} + \frac{3}{4}$, and all fallback nodes are unactivated and have score 1. Thus, after the n_H selector tails, the next ℓ highest-scoring nodes are precisely the ℓ selected red candidates. Consequently,

$$|T_k(G \oplus \Delta) \cap V_{\text{red}}| = \ell,$$

and hence

$$\text{Unfair}(T_k(G \oplus \Delta)) = 0 < \varepsilon.$$

Therefore

$$\text{OPT}_\varepsilon \leq \ell.$$

NO case. Suppose H contains no independent set of size ℓ . We show that no feasible edit set of size smaller than Q exists.

Let Δ be any edit set with $|\Delta| < Q$. Since each fallback node requires all Q of its private fallback edges to reach score $\frac{Q}{2} + 1$, no fallback node is activated by Δ . Indeed, every fallback node has score at most

$$\frac{Q}{2} + \frac{1}{2} < \frac{Q}{2} + \frac{3}{4},$$

which is below the blue buffer score.

Let

$$S_\Delta := \{u_i \in U : t_i \rightarrow r_i \in \Delta\}$$

be the set of core vertices selected by Δ .

If $|S_\Delta| < \ell$, then fewer than ℓ red core candidates have score above the buffer nodes, and no fallback red node has score above the buffers. Thus, among the ℓ top- k positions below the selector tails, fewer than ℓ are red. Hence the top- k set does not contain exactly ℓ red nodes.

Now suppose $|S_\Delta| \geq \ell$. Since H has no independent set of size ℓ , the set S_Δ cannot be independent. Therefore there exists an edge $e = \{u_i, u_j\} \in F$ with both endpoints in S_Δ . For this edge e , all ℓ conflict clones

$$q_{e,1}, \dots, q_{e,\ell}$$

have score

$$\frac{Q}{2} + \frac{5}{2},$$

which is strictly larger than the score $\frac{Q}{2} + \frac{3}{2}$ of every selected red candidate. After the n_H selector tails, the next ℓ highest-scoring nodes are these ℓ blue conflict clones. Thus the top- k set again does not contain exactly ℓ red nodes.

In both cases, Δ is infeasible. Since Δ was arbitrary with $|\Delta| < Q$, every feasible solution in the NO case has size at least Q :

$$\text{OPT}_\varepsilon \geq Q.$$

Feasibility of the NO instances. The constructed dual instance is always feasible. Indeed, by adding all Q private fallback edges into each of the ℓ fallback nodes, each fallback node obtains score

$$\frac{Q}{2} + 1 > \frac{Q}{2} + \frac{3}{4},$$

which is above the buffer score. If no core selector edges are added, then all conflict clones have score 2, and the ℓ activated fallback nodes occupy the ℓ positions below the n_H selector tails. Therefore the resulting top- k set contains exactly ℓ red nodes. Hence $\text{OPT}_\varepsilon < \infty$ for every constructed instance.

Gap and approximation contradiction. We have shown:

$$\text{YES instance:} \quad \text{OPT}_\varepsilon \leq \ell,$$

whereas

$$\text{NO instance:} \quad \text{OPT}_\varepsilon \geq Q.$$

By construction, $Q > \psi\ell$.

Now suppose there were a polynomial-time ψ -approximation algorithm for the dual problem. On a YES instance, it would return a feasible edit set of size at most

$$\psi \text{OPT}_\varepsilon \leq \psi\ell < Q.$$

On a NO instance, every feasible edit set has size at least Q , so the algorithm must return a solution of size at least Q . Thus, by checking whether the returned solution has size smaller than Q , we could decide whether H contains an independent set of size at least ℓ .

This would solve INDEPENDENT SET on 3-regular graphs in polynomial time, contradicting $P \neq NP$. Therefore no polynomial-time ψ -approximation exists unless $P = NP$. $\square$

Proof of Corollary 3.3. We assume rational input parameters with polynomial bit complexity; Katz scores can then be computed exactly by solving a rational linear system, and score comparisons can be performed in polynomial time. The problem is in NP: given an edit set Δ , we can check in polynomial time whether $|\Delta| \leq C$ and whether the edited graph satisfies the target fairness constraint. NP-hardness follows from the gap construction in Theorem 3.2. In particular, distinguishing whether $\text{OPT}_\varepsilon \leq \ell$ or $\text{OPT}_\varepsilon \geq Q$ is NP-hard, and therefore so is the threshold decision problem. $\square$

Proof of Proposition 4.1. We have $K_G = \sum_{k=0}^{\infty} \alpha^k A_G^k$ and $K_{G'} = \sum_{k=0}^{\infty} \alpha^k A_{G'}^k$. Adding the edge $u \rightarrow v$ yields $A_{G'} = A_G + E$, where E is the matrix with a single 1 in position (u, v) and zeros elsewhere. Hence $A_{G'} \geq A_G$ entrywise. For nonnegative matrices, entrywise inequality implies $\rho(A_{G'}) \geq \rho(A_G)$, so $\alpha < 1/\rho(A_{G'}) \leq 1/\rho(A_G)$ and both Neumann series converge. We prove $A_{G'}^k \geq A_G^k$ entrywise for all $k \geq 0$ by induction. For $k = 0$, both are the identity matrix. Assuming $A_{G'}^k \geq A_G^k$ entrywise,

$$A_{G'}^{k+1} = A_{G'} \cdot A_{G'}^k \geq A_G \cdot A_{G'}^k \geq A_G \cdot A_G^k = A_G^{k+1},$$

where the first inequality uses $A_{G'} \geq A_G$ and the second uses the induction hypothesis. Multiplying by $\alpha^k \geq 0$ and summing over k yields $K_{G'} \geq K_G$ entrywise. $\square$

Proof of Lemma 4.2. Adding $u \rightarrow v$ gives $A_{G'} = A_G + \mathbf{e}_u \mathbf{e}_v^\top$. By Sherman–Morrison [1] with $M = I - \alpha A_G$, $\mathbf{x} = \alpha \mathbf{e}_u$, $\mathbf{y} = \mathbf{e}_v$:

$$\begin{aligned} K_{G'} &= (M - \mathbf{x} \mathbf{y}^\top)^{-1} = K_G + \frac{\alpha K_G \mathbf{e}_u \mathbf{e}_v^\top K_G}{1 - \alpha \mathbf{e}_v^\top K_G \mathbf{e}_u} \\ &= K_G + \frac{\alpha}{1 - \alpha K_G(v, u)} \mathbf{k}_u \mathbf{k}_v. \end{aligned}$$

The denominator is positive: since $\alpha < 1/\rho(A_{G'})$, the rank-one matrix $\mathbf{k}_u \mathbf{k}_v$ is nonnegative by Proposition 4.1, so $\alpha/(1 - \alpha K_G(v, u)) > 0$, giving $1 - \alpha K_G(v, u) > 0$.

Finally, we have $\Delta s(a \mid u \rightarrow v) = \mathbf{1}^\top (K_{G'} - K_G) \mathbf{e}_a = \frac{\alpha}{1 - \alpha K_G(v, u)} (\mathbf{1}^\top \mathbf{k}_u) (\mathbf{k}_v \cdot \mathbf{e}_a) = \frac{\alpha s_G(u) K_G(v, a)}{1 - \alpha K_G(v, u)}$. $\square$

Proof of Proposition 4.4. Apply Equation (3) to each term in Definition 4.3 and factor. $\lambda_G(u, v) > 0$ since $\alpha > 0$, $s_G(u) > 0$, and $1 - \alpha K_G(v, u) > 0$ under the Katz-validity assumption. $\square$

Proof of Proposition 5.1. The two identities follow directly from the single-edge Katz update formula with head node a . Katz validity implies nonnegativity of the kernel, $K_G(a, a) \geq 1$, $K_G(a, u) \geq 0$, and $1 - \alpha K_G(a, u) > 0$. Hence $\frac{K_G(a, a)}{1 - \alpha K_G(a, u)} \geq 1$, so the proxy does not overestimate the direct gain of a . The sign of the differential gain follows from the second identity because $\hat{g}_G(u, a) > 0$ and the denominator is positive. $\square$

POPULATION-PROPORTIONAL TARGETS

Katz methods use the Katz ranking, whereas the PageRank-oriented methods use their resulting PageRank ranking; consequently, their initial top- k counts can differ. PageRank-oriented methods do not add edges, so their edit counts are marked “–.” OOT denotes the one-hour limit, and OOM denotes termination due to insufficient memory. Penn GAPGREEDY was stopped after 2,438 seconds, and Pokec GAPGREEDY was not started after three consecutive one-hour timeouts; both are marked OOT.

TABLE I: Population-proportional target results on Blogs.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
52	BLADE	32	48	52	Yes	0.0000	0.96	0.004
52	SAMEGROUP	1596	48	52	Yes	0.0000	0.93	0.224
52	KATZMASS	820	48	49	No	0.0009	0.99	0.011
52	GAPGREEDY	36	48	52	Yes	0.0000	0.96	12.245
52	FAIRGD	–	48	50	No	0.0004	0.98	0.170
52	FAIRWALK	–	48	47	No	0.0025	0.86	0.000
52	CROSSWALK	–	48	47	No	0.0025	0.68	0.009
52	LFPR-N	–	48	56	No	0.0016	0.80	0.018
52	LFPR-U	–	48	54	No	0.0004	0.90	0.037

TABLE II: Population-proportional target results on Hopkins.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
55	BLADE	228	44	55	Yes	0.0000	0.89	0.339
55	SAMEGROUP	2998	44	55	Yes	0.0000	0.89	4.887
55	KATZMASS	83402	44	55	Yes	0.0000	0.89	1.062
55	GAPGREEDY	OOT	44	OOT	OOT	OOT	OOT	OOT
55	FAIRGD	–	53	56	No	0.0001	0.97	1.745
55	FAIRWALK	–	53	58	No	0.0009	0.93	0.002
55	CROSSWALK	–	53	60	No	0.0025	0.70	0.022
55	LFPR-N	–	53	60	No	0.0025	0.87	0.110
55	LFPR-U	–	53	55	Yes	0.0000	0.89	0.948

TABLE III: Population-proportional target results on Retweet.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
61	BLADE	516	34	61	Yes	0.0000	0.73	1.093
61	SAMEGROUP	4562	34	61	Yes	0.0000	0.71	0.819
61	KATZMASS	100940	34	20	No	0.1681	0.85	6.045
61	GAPGREEDY	OOT	34	OOT	OOT	OOT	OOT	OOT
61	FAIRGD	–	48	48	No	0.0169	0.99	0.743
61	FAIRWALK	–	48	57	No	0.0016	0.90	0.001
61	CROSSWALK	–	48	52	No	0.0081	0.33	0.011
61	LFPR-N	–	48	74	No	0.0169	0.72	8.349
61	LFPR-U	–	48	70	No	0.0081	0.77	9.906

TABLE IV: Population-proportional target results on Deezer.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
44	BLADE	117	26	44	Yes	0.0000	0.82	0.606
44	SAMEGROUP	1562	26	44	Yes	0.0000	0.77	5.842
44	KATZMASS	30318	26	23	No	0.0441	0.97	1.197
44	GAPGREEDY	OOT	26	OOT	OOT	OOT	OOT	OOT
44	FAIRGD	–	34	34	No	0.0100	1.00	1.342
44	FAIRWALK	–	34	30	No	0.0196	0.94	0.002
44	CROSSWALK	–	34	41	No	0.0009	0.67	0.029
44	LFPR-N	–	34	34	No	0.0100	0.68	10.394
44	LFPR-U	–	34	35	No	0.0081	0.74	25.355

TABLE V: Population-proportional target results on Penn.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
48	BLADE	3666	20	48	Yes	0.0000	0.72	23.366
48	SAMEGROUP	16388	20	45	No	0.0009	0.75	82.865
48	KATZMASS	1400650	20	35	No	0.0169	0.85	26.565
48	GAPGREEDY	OOT	20	OOT	OOT	OOT	OOT	OOT
48	FAIRGD	–	58	58	No	0.0100	1.00	50.198
48	FAIRWALK	–	58	53	No	0.0025	0.93	0.023
48	CROSSWALK	–	58	47	No	0.0001	0.68	0.237
48	LFPR-N	–	58	45	No	0.0009	0.72	6.236
48	LFPR-U	–	58	48	Yes	0.0000	0.75	70.913

TABLE VI: Population-proportional target results on Pokec.

Target	Algorithm	Edits	Initial $ T_k \cap V_r $	Final $ T_k \cap V_r $	Success	Unfair(T_k)	Overlap@100	Time (s)
49	BLADE	512	38	49	Yes	0.0000	0.89	84.532
49	SAMEGROUP	2559	38	49	Yes	0.0000	0.89	10.324
49	KATZMASS	OOT	38	OOT	OOT	OOT	OOT	OOT
49	GAPGREEDY	OOT	38	OOT	OOT	OOT	OOT	OOT
49	FAIRGD	–	53	53	No	0.0016	1.00	1889.701
49	FAIRWALK	–	53	60	No	0.0121	0.87	0.844
49	CROSSWALK	–	53	58	No	0.0081	0.85	232.533
49	LFPR-N	OOM	53	OOM	OOM	OOM	OOM	OOM
49	LFPR-U	OOM	53	OOM	OOM	OOM	OOM	OOM

REFERENCES

- [1] J. Sherman and W. J. Morrison, “Adjustment of an inverse matrix corresponding to a change in one element of a given matrix,” *The Annals of Mathematical Statistics*, vol. 21, no. 1, pp. 124–127, 1950.